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SYSTEMS AND METHODS FOR A THERMAL MANAGEMENT CONTROL VALVE WITH A ROTATING LOUVER ASSEMBLY

Abstract

A thermal management control valve a housing assembly having a first housing section and a second housing section, a rotary actuator coupled to a drive shaft, and a louver plate assembly. The louver plate assembly includes a louver plate having a louver cutout, a first seal assembly arranged between the louver plate and the first housing section, and a second seal assembly arranged between the louver plate and the second housing section. The first seal assembly includes a plurality of inner walls that define a plurality of first fluid passageways. The second seal assembly defines a second fluid passageway between. Selective rotation of the louver plate rotationally aligns the louver cutout with one or more of the plurality of first fluid passageways and the second fluid passageway to provide fluid communication between the one or more of the plurality of first fluid passageways and the second fluid passageway.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application claims the benefit of and priority to U.S. Provisional Application No. 63/560,550, filed on Mar. 1, 2024, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND
[0002] Thermal control valves are used, for example, in hybrid and battery-powered vehicle applications to control a flow of working fluid that acts to heat or cool a device or component.
SUMMARY
[0003] In some aspects, the present disclosure relates to a thermal management control valve including: a housing assembly including a first housing section and a second housing section; a rotary actuator coupled to a drive shaft that extends into the housing assembly; and a louver plate assembly enclosed within the housing assembly and including: a louver plate including a louver cutout extending through the louver plate, wherein the louver plate is coupled to the drive shaft so that selective rotation of the drive shaft is configured to rotate the louver plate; a first seal assembly arranged between the louver plate and the first housing section, wherein the first seal assembly includes a plurality of inner walls that define a plurality of first fluid passageways between the louver plate and the first housing section; and a second seal assembly arranged between the louver plate and the second housing section, wherein the second seal assembly defines a second fluid passageway between the louver plate and the second housing section, wherein selective rotation of the louver plate rotationally aligns the louver cutout with one or more of the plurality of first fluid passageways and the second fluid passageway to provide fluid communication between the one or more of the plurality of first fluid passageways and the second fluid passageway.
[0004] In some aspects, the present disclosure relates to a thermal management control valve including: a housing assembly including a first housing section and a second housing section, wherein the first housing section includes a plurality of first fluid ports and the second housing section includes a plurality of second fluid ports; a rotary actuator coupled to a drive shaft that extends into the housing assembly; and a louver plate assembly enclosed within the housing assembly and including: a louver plate including a louver cutout extending through the louver plate, wherein the louver plate is coupled to the drive shaft so that selective rotation of the drive shaft is configured to rotate the louver plate; a first seal assembly sealed between the louver plate and each of the plurality of first fluid ports; and a second seal assembly sealed between the louver plate and each of the plurality of second fluid ports, wherein selective rotation of the louver plate rotationally aligns the louver cutout with one or more of the plurality of first fluid ports and one or more of the plurality of second fluid ports to provide fluid communication between the one or more of the plurality of first fluid ports and the one or more of the plurality of second fluid ports.
[0005] In some aspects, the present disclosure relates to a thermal management control valve including: a housing assembly including a first housing section and a second housing section; a rotary actuator coupled to a drive shaft that extends into the housing assembly; a first louver plate assembly enclosed within the housing assembly and including a first louver plate arranged about a first central axis; and a second louver plate assembly enclosed within the housing assembly and including a second louver plate arranged about a second central axis that is spaced from the first

central axis, wherein the first louver plate assembly and the second louver plate assembly both include a first seal plate assembly and a second seal plate assembly, wherein the first seal plate assembly of the first louver plate assembly is sealed between the first louver plate and the first housing section, and the second seal plate assembly of the first louver plate assembly is sealed between the first louver plate and the second housing section, wherein the first seal plate assembly of the second louver plate assembly is sealed between the second louver plate and the first housing section, and the second seal plate assembly of the second louver plate assembly is sealed between the second louver plate and the second housing section, and wherein selective rotation of the drive shaft is configured to rotate the first louver plate and the second louver plate in opposing directions.
[0006] This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0007] The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:
[0008] FIG. **1** is a perspective view of a thermal management control valve, according to an exemplary embodiment;
[0009] FIG. **2** is an exploded view of the thermal management control valve of FIG. **1**;
[0010] FIG. **3** is a top view of the thermal management control valve of FIG. **1**;
[0011] FIG. **4** is a bottom view of the thermal management control valve of FIG. **1**;
[0012] FIG. **5** is a perspective view of a first seal assembly of a first louver assembly of the thermal management control valve of FIG. **1**;
[0013] FIG. **6** is a top view of the first seal assembly of FIG. **5**;
[0014] FIG. **7** is a perspective view of a second seal assembly of a first louver assembly of the thermal management control valve of FIG. **1**;
[0015] FIG. **8** is a top view of the second seal assembly of FIG. **7**;
[0016] FIG. **9** is a perspective view of a first seal assembly of a second louver assembly of the thermal management control valve of FIG. **1**;
[0017] FIG. **10** is a top view of the first seal assembly of FIG. **9**;
[0018] FIG. **11** is a perspective view of a second seal assembly of a second louver assembly of the thermal management control valve of FIG. **1**;
[0019] FIG. **12** is a top view of the second seal assembly of FIG. **11**;
[0020] FIG. **13** is a bottom perspective view of a first housing section of a first louver assembly of the thermal management control valve of FIG. **1**;
[0021] FIG. **14** is a top perspective view of a first louver assembly of the thermal management control valve of FIG. **1**;
[0022] FIG. **15** is a top view of a first louver assembly of the thermal management control valve of FIG. **1**, with a first housing section removed;
[0023] FIG. **16** is a top view of a first louver assembly of the thermal management control valve of FIG. **1**, with a first housing section and a first seal assembly of both a first louver assembly and a second louver assembly removed;
[0024] FIG. **17** is a cross-sectional view of a first louver assembly of the thermal management control valve of FIG. **1**;
[0025] FIG. **18** is a perspective view of a first louver assembly of the thermal management control

valve of FIG. **1**, including a plurality of fluid ports;
[0026] FIG. **19** is a top cross-sectional view of the thermal management control valve of FIG. **18**;
[0027] FIG. **20** is a bottom cross-sectional view of the thermal management control valve of FIG. **18**;
[0028] FIG. **21** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a first operating mode;
[0029] FIG. **22** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a first intermediate operating mode;
[0030] FIG. **23** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a second operating mode;
[0031] FIG. **24** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a third operating mode;
[0032] FIG. **25** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a third intermediate operating mode;
[0033] FIG. **26** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a fourth operating mode;
[0034] FIG. **27** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a fourth intermediate operating mode;
[0035] FIG. **28** is a top view of a flow control assembly of the thermal management control valve of FIG. **1** in a fifth operating mode;
[0036] FIG. **29** is a perspective view of a thermal management control valve, according to an exemplary embodiment;
[0037] FIG. **30** is an exploded view of the thermal management control valve of FIG. **29** with a rotary actuator removed;
[0038] FIG. **31** is a top view of the thermal management control valve of FIG. **29** with a rotary actuator removed;
[0039] FIG. **32** is a bottom perspective view of the thermal management control valve of FIG. **29** with a rotary actuator removed;
[0040] FIG. **33** is a bottom view of the thermal management control valve of FIG. **29** in a first position;
[0041] FIG. **34** is a bottom view of the thermal management control valve of FIG. **29** in a first intermediate position;
[0042] FIG. **35** is a bottom view of the thermal management control valve of FIG. **29** in a second intermediate position;
[0043] FIG. **36** is a bottom view of the thermal management control valve of FIG. **29** in a second position; and
[0044] FIG. **37** is a cross-sectional view of the thermal management control valve of FIG. **29** within a housing assembly.

DETAILED DESCRIPTION

[0045] Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.
[0046] The use herein of the term “axial” and variations thereof refers to a direction that extends generally along an axis of symmetry, a central axis, or an elongate direction of a particular component or system. For example, axially extending features of a component may be features that extend generally along a direction that is parallel to an axis of symmetry or an elongate direction of that component. Similarly, the use herein of the term “radial” and variations thereof refers to directions that are generally perpendicular to a corresponding axial direction. For example, a

radially extending structure of a component may generally extend at least partly along a direction that is perpendicular to a longitudinal or central axis of that component. The use herein of the term “circumferential” and variations thereof refers to a direction that extends generally around a circumference of an object or around an axis of symmetry, a central axis or an elongate direction of a particular component or system.

[0047] Hybrid electric and battery electric vehicles include a high-voltage battery or an array of high-voltage battery packs that supply electric power to various components on the vehicle (e.g., drive motors, user interfaces, power take off (PTO) units, electric power steering motors, etc.). The automotive vehicle market (e.g., on-road or on-highway vehicles) is trending toward the development of longer all-electric ranges to allow a user to drive longer distances between battery charges. The demand for longer all-electric range requires batteries/battery packs with higher capacities that define a larger mounting footprint/volume, when compared to lower all-electric range vehicles. Also, increasing the battery/battery pack capacity typically increases the thermal management demand (e.g., heating and cooling capacities) required to provide temperature control of the battery/battery pack and the various other components on the vehicle that are actively or passively heated and/or cooled (e.g., power electronics, drive motors, HVAC systems etc.). So there is less space available to mount thermal management components (e.g., control valves, pumps, heat exchangers, conduits/tubing) and a need for the thermal management components to meet higher requirements for heating and cooling capacities, which is conventionally accommodated by using larger thermal management components.

[0048] The systems and methods of the present disclosure relate to a thermal management control valve that includes a flow control assembly (e.g., a louver plate assembly) that is selectively rotated to open/close various flow paths that within a housing of the thermal management control valve. The thermal management control valve includes a louver plate assembly having a louver plate, a first seal plate assembly arranged on a first side of the louver plate, and a second seal plate assembly arranged on a second side of the louver plate. In general, the first seal plate assembly defines one or more first sealed fluid channels or passageways between the louver plate and a corresponding number of first fluid ports, and the second seal plate assembly defines one or more second sealed fluid channels or passageways between the louver plate and a corresponding number of second fluid ports. The selective rotation of the louver plate aligns a cutout or slot formed in the louver plate with various combinations of the one or more first sealed fluid passageways and the one or more second sealed fluid passageways, which, in turn, provides fluid communication between various combinations of the first fluid ports and the second fluid ports. In this way, for example, the louver plate assembly is able service several flow paths and components (e.g., pumps, valves, heat exchangers) that receive fluid flow from the flow fluid ports, while maintaining a small packaging size (e.g., when compared to conventional thermal management control valves)

[0049] FIGS. **1-4** show a thermal management control valve **100**, according to an exemplary embodiment of the present disclosure. In general, the thermal management control valve **100** is configured to control fluid flow (e.g., direction, routing, flowrate, and/or on/off control) for one or more working fluids in heating and/or cooling systems on an electric vehicle. The use of the term “electric vehicle” and variations thereof refers to a battery electric vehicle, a fully electric vehicle, a plug-in-hybrid vehicle, or a hybrid vehicle that is intended to be driven on a road or highway. In some embodiments, the thermal management control valve **100** provides flow control for a single type of working fluid (e.g., air, water, refrigerant, oil, etc.).

[0050] The thermal management control valve **100** includes a housing, shown as housing assembly **102**, and a flow control assembly **104** enclosed within the housing assembly **102**. The housing assembly **102** includes a first or upper housing section **106** and a second or lower housing section **108**. In some embodiments, the first housing section **106** and the second housing section **108** are fabricated from a polymer or plastic material. In some embodiments, the first housing section **106** and the second housing section **108** are manufactured via a plastic injection molding.

[0051] In general, the coupling formed between the first housing section **106** and the second housing section **108** is formed along a coplanar interface. For example, a bonding interface **110** formed between the first housing section **106** and the second housing section **108** is formed between coplanar surfaces of the first housing section **106** and the second housing section **108**. Forming the bonding interface **110** between coplanar surfaces forms coplanar bonding interfaces that enable a simplified and efficient bonding processes to be utilized to form the bonds between the first housing section **106** and the second housing section **108**. Additionally, forming the bonds between the first housing section **106** and the second housing section **108** along a coplanar bonding interface improves the manufacturing efficiency of the housing assembly **102**.

[0052] Each of the first housing section **106** and the second housing section **108** includes one or more ports that facilitate a connection between the housing assembly **102** and a thermal management component (e.g., a heat exchanger, a pump, a check valve, a conduit/tube, etc.) or another port. In general, the housing assembly **102** may be designed to include any port configuration (e.g., number and arrangement of the ports) dictated by a particular electric vehicle application. In some embodiments, the port configuration may be quickly and efficiently changed by modifying a mold design used to plastic injection mold the housing assembly **102**, and/or by modifying the structure of the flow control assembly **104** as described herein.

[0053] In the exemplary embodiment, the flow control assembly **104** includes a first louver plate assembly **112** and a second louver plate assembly **114**. Each of the first louver plate assembly **112** and the second louver plate assembly **114** is enclosed between the first housing section **106** and the second housing section **108** and is configured to selectively control fluid flow between a plurality of ports formed in the housing assembly **102**. The inclusion of two louver plate assemblies in the thermal management control valve **100** is but one exemplary embodiment, and the thermal management control valve **100** may include a single louver plate assembly, or more than two louver plate assemblies, and still function to control fluid flow between a plurality of ports formed in the housing assembly **102**, or other ports formed external to or in fluid communication with the housing assembly **102**.

[0054] The first housing section **106** includes one or more ports, each defining a cutout, aperture, channel, passageway, or through hole, that allow fluid flow into or out of the housing assembly **102**. In the exemplary embodiment, the first housing section **106** includes a first housing port **116**, a second housing port **118**, a third housing port **120**, a fifth housing port **124**, and a sixth housing port **126**. The first housing port **116**, the second housing port **118**, and the third housing port **120** are arranged generally above (e.g., from the perspective of FIG. **2**) and is in fluid communication with the first louver plate assembly **112**, and the fifth housing port **124** and the sixth housing port **126** are arranged generally above (e.g., from the perspective of FIG. **2**) and are in fluid communication with the second louver plate assembly **114**. In some embodiments, the first housing port **116**, the second housing port **118**, the third housing port **120**, the fifth housing port **124**, and the sixth housing port **126** may each be in fluid communication with an individual connector, fitting, or tubing that delivers fluid flow to or from a thermal management component. In some embodiments, two or more of the first housing port **116**, the second housing port **118**, the third housing port **120**, the fifth housing port **124**, and the sixth housing port **126** may be in fluid communication with a common connector, fitting, or tubing that delivers fluid flow to or from a thermal management component. For example, in an exemplary embodiment, the second housing port **118** and the fifth housing port **124** may be in fluid communication with a common connector, fitting, or tubing, and the third housing port **120** and the sixth housing port **126** may be in fluid communication with a common connector, fitting, or tubing.

[0055] The second housing section **108** includes one or more ports, each defining a cutout, aperture, channel, passageway, or through hole, that allow fluid flow into or out of the housing assembly **102**. In the exemplary embodiment, the second housing section **108** includes a fourth housing port **122**, a seventh housing port **128**, and an eighth housing port **130**. The fourth housing port **122** is arranged generally below (e.g., from the perspective of FIG. **2**) and is in fluid communication with the first louver plate assembly **112**, and the seventh housing port **128** and the eighth housing port **130** are arranged generally below (e.g., from the perspective of FIG. **2**) and are in fluid communication with the second louver plate assembly **114**. In some embodiments, each of the fourth housing port **122**, the seventh housing port **128**, and the eighth housing port **130** may each be in fluid communication with an individual connector, fitting, or tubing that delivers fluid flow to or from a thermal management component. In some embodiments, two or more of the fourth housing port **122**, the seventh housing port **128**, and the eighth housing port **130** may be in fluid communication with a common connector, fitting, or tubing that delivers fluid flow to or from a thermal management component.

[0056] In general, the flow control assembly **104** is configured to selectively control fluid flow between the various ports of the housing assembly **102** (e.g., the first housing port **116**, the second housing port **118**, the third housing port **120**, the fifth housing port **124**, the sixth housing port **126**, the fourth housing port **122**, the seventh housing port **128**, and the eighth housing port **130**). In the exemplary embodiment, the flow control assembly **104** includes a rotary actuator **132** (e.g., a rotary motor, a stepper motor, an electric motor, etc.) that is coupled to each of the first louver plate assembly **112** and the second louver plate assembly **114**. The rotary actuator **132** is coupled to a drive shaft **134** by a gear train **136**. In some embodiments, the rotary actuator **132** may be directly coupled to the drive shaft **134** without the gear train **136** coupled between the drive shaft **134** and the rotary actuator **132**.

[0057] During operation, the rotary actuator **132** is configured to rotate in a predetermined direction and a predetermined magnitude (e.g., a predetermined rotational distance) in response to energization of the rotary actuator **132**. The rotary actuator **132** defines a drive axis **138** and extends into a louver cavity or chamber, shown as a louver manifold **140** formed within the housing assembly **102**. For example, a portion of the louver manifold **140** may be formed by the first housing section **106** and a portion of the louver manifold **140** may be formed by the second housing section **108**. When the first housing section **106** is installed on the second housing section **108**, the respective portions of the louver manifold **140** formed by the first housing section **106** and the second housing section **108** define an enclosure that forms the louver manifold **140** and houses the components of the first louver plate assembly **112** and the second louver plate assembly **114**.

[0058] The drive shaft **134** extends into the housing assembly **102** (e.g., into the second housing section **108**) at a location that is offset (e.g., does not intersect with) from the bonding interface **110** formed between the first housing section **106** and the second housing section **108**. In other words, an entirety of the drive shaft **134** is offset from the bonding interface **110**. This offset arrangement of the drive shaft **134** enables the bonding interface **110** to maintain its coplanar orientation and the associated manufacturing advantages associated therewith. For example, the bonding interface **110** is not required to form a seal around a split interface that curves around opposing sides of the drive shaft **134**.

[0059] In general, the first louver plate assembly **112** includes similar components as the second louver plate assembly **114** and both the first louver plate assembly **112** and the second louver plate assembly **114** provide similar functionality within the thermal management control valve **100**. The individual components within the first louver plate assembly **112** and the second louver plate assembly **114** may not be identically designed, but the type, function, and number of components may be similar between the first louver plate assembly **112** and the second louver plate assembly **114**. The components of the first louver plate assembly **112** described herein include the suffix "a" and the components of the second louver plate assembly **114** described herein include the suffix "b."

[0060] In the exemplary embodiment, the first louver plate assembly **112** includes a first louver plate **150***a*, a first seal assembly **152***a* (e.g., an upper seal assembly from the perspective of FIG. **2**), and a second seal assembly **154***a* (e.g., a lower seal assembly from the perspective of FIG. **2**). The first louver plate assembly **112** is arranged along a first central axis **156***a*, which is defined along a central rod or pin **158***a*. The first or upper seal assembly **152***a* is arranged between the first louver plate **150***a* and the first housing section **106** (e.g., in a direction along the first central axis **156***a*, and the second seal assembly **154***a* is arranged between the first louver plate **150***a* and the second housing section **108**. In other words, the first seal assembly **152***a* and the second seal assembly **154***a* are arranged on opposing sides of the first louver plate **150***a*.

[0061] The first louver plate **150***a* defines a generally circular or round shape and includes a plurality of gear teeth arranged around an outer perimeter of the first louver plate **150***a*. A cutout, aperture, hole, slot, shown as louver cutout **160***a* extends through the first louver plate **150***a*. Specifically, the louver cutout **160***a* extends through both an upper or first louver surface **162***a* and a lower or second louver surface **164***a*. In the exemplary embodiment, the louver cutout **160***a* defines a generally triangular shape, with two straight edges and one convex or rounded edge. In some embodiments, the louver cutout **160***a* may be shaped differently (e.g., round, circular, rectangular, or another shape) to provide the appropriate connections between the sealed fluid channels or fluid passageways formed within the first louver plate assembly **112**.

[0062] In general, the first seal assembly **152***a* is sealed between the first louver plate **150***a* and the first housing section **106** and defines one or more fluid channels or fluid passageways, each extending between the first louver plate **150***a* and one of the first housing port **116**, the second housing port **118**, and the third housing port **120**. With specific reference to FIGS. **2**, **5**, and **6**, in the exemplary embodiment, the first seal assembly **152***a* includes a first static seal plate **166***a*, a first seal plate **168***a*, and a first static seal **170***a*. When the first louver plate assembly **112** is assembled, the first static seal plate **166***a* is in engagement with and sealed against the upper or first louver surface **162***a* of the first louver plate **150***a* (see, e.g., FIG. **17**). The first seal plate **168***a* is arranged between the first static seal plate **166***a* and the first static seal **170***a* (e.g., in a direction along the first central axis **156***a*), and is sealed against both the first static seal plate **166***a* and the first static seal **170***a*.

[0063] The first static seal **170***a* forms a seal between the first housing section **106** and the first seal plate **168***a*. In some embodiments, the first static seal **170***a* is fabricated from an elastomer material. In the exemplary embodiment, the first static seal **170***a* is compressed between the first housing section **106** and the first seal plate **168***a*, so that the first static seal **170***a* provides a biasing force on the first seal plate **168***a* and the first static seal plate **166***a* in a direction toward the first louver plate **150***a*, which aids in maintaining a seal between the first louver plate **150***a* and the first housing section **106**.

[0064] In general, each of the first static seal plate **166***a*, the first seal plate **168***a*, and the first static seal **170***a* include similar features that are arranged in the same rotational orientation about the first central axis **156***a*, which aid in defining one or more of the sealed fluid channels or passageways between the first louver plate **150***a* and the first housing section **106**. Specifically, each of the first static seal plate **166***a*, the first seal plate **168***a*, and the first static seal **170***a* includes a first outer wall **172***a*, a first inner wall **174***a*, a first central hub **176***a*, and a plurality of partitions, walls, or dividers, shown as first inner walls **178***a*. Each of the first inner walls **178***a* extends radially (e.g., relative to the first central axis **156***a*) between the first inner wall **174***a* and the first central hub **176***a*.

[0065] In general, a number of the first inner walls **178***a* within the first seal assembly **152***a* defines the number of sealed fluid channels or passageways formed within the first seal assembly **152***a* (e.g., one or less of the inner walls defines one fluid passageway, two of the inner walls defines two fluid passageways, three of the inner walls defines three fluid passageways, etc.). In the exemplary embodiment, the first seal assembly **152***a* includes three of the first inner walls **178***a*, which divides the flow area between the first louver plate **150***a* and the first housing section **106** into a first fluid passageway **180***a*, a second fluid passageway **182***a*, and a third fluid passageway **184***a*. The first fluid passageway **180***a* extends between the first louver plate **150***a* and the first housing port **116**, the second fluid passageway **182***a* extends between the first louver plate **150***a* and the second housing port **118**, and the third fluid passageway **184***a* extends between the first louver plate **150***a* and the third housing port **120**. In some embodiments, the first seal assembly **152***a* may include more or less than three of the first inner walls **178***a* to define more or less than three fluid passageways to accommodate a particular porting configuration in the first housing section **106**.

[0066] In the exemplary embodiment, each of the first inner walls **178***a* is rotationally or circumferentially spaced from one another. The circumferential spacing between adjacent pairs of the first inner walls **178***a* defines circumferential length of each of the first fluid passageway **180***a*, the second fluid passageway **182***a*, and the third fluid passageway **184***a* (e.g., a rotational angle about the first central axis **156***a* over which the louver cutout **160***a* may travel to allow fluid flow through the respective passageways formed in the first seal assembly **152***a*). In some embodiments, the circumferential length of two or more the first fluid passageway **180***a*, the second fluid passageway **182***a*, and the third fluid passageway **184***a* may be different. For example, a circumferential length **186***a* of the first fluid passageway **180***a* may be different than a circumferential length **188***a* of the second fluid passageway **182***a* and/or different than a circumferential length **190***a* of the third fluid passageway **184***a*. In some embodiments, the circumferential length of two or more of the first fluid passageway **180***a*, the second fluid passageway **182***a*, and the third fluid passageway **184***a* may approximately equal. For example, the circumferential length **188***a* of the second fluid passageway **182***a* may be approximately equal to the circumferential length **190***a* of the third fluid passageway **184***a*.

[0067] In the exemplary embodiment, the first static seal plate **166***a* and the first seal plate **168***a* may include one more anti-rotation features that rotationally fix the first static seal plate **166***a* and the first seal plate **168***a* relative to the housing assembly **102** (i.e., prevent the first static seal plate **166***a* and the first seal plate **168***a* from rotating relative to the housing assembly **102**). For example, the first static seal plate **166***a* includes a lip **192***a* that protrudes outwardly from the first outer wall **172***a* of the first static seal plate **166***a*. In general, the lip **192***a* interrupts the round or circular shape of the first outer wall **172***a* of the first static seal plate **166***a*, and includes a pair of anti-rotation surfaces **194***a* arranged at opposing edges of the lip **192***a*. The first static seal plate **166***a* also includes a recess or notch **196***a* formed in the lip **192***a* between the anti-rotation surfaces **194***a*.

[0068] The first seal plate **168***a* includes a plurality of tabs **198***a* that extend outwardly from the first outer wall **172***a* of the first seal plate **168***a*. One of the plurality of tabs **198***a* is received within the notch **196***a* of the first static seal plate **166***a*, which aids in preventing relative rotation between the first static seal plate **166***a* and the first seal plate **168***a*. As described herein, the plurality of tabs **198***a* and/or the anti-rotation surfaces **194***a* may engage a structure within the housing assembly **102** to prevent relative rotation between the housing assembly **102** and both the first static seal plate **166***a* and the first seal plate **168***a*.

[0069] In general, the second seal assembly **154***a* is sealed between the first louver plate **150***a* and the second housing section **108** and defines one or more fluid channels or fluid passageways, each extending between the first louver plate **150***a* and the fourth housing port **122**. With specific reference to FIGS. **2**, **7**, and **8**, in the exemplary embodiment, the second seal assembly **154***a* includes a second static seal plate **200***a*, a second seal plate **202***a*, and a second static seal **204***a*. When the first louver plate assembly **112** is assembled, the second static seal plate **200***a* is in engagement with and sealed against the lower or second louver surface **164***a* of the first louver plate **150***a* (see, e.g., FIG. **17**). The second seal plate **202***a* is arranged between the second static seal plate **200***a* and the second static seal **204***a* (e.g., in a direction along the first central axis **156***a*), and is sealed against both the second static seal plate **200***a* and the second static seal **204***a*.

[0070] The second static seal **204***a* forms a seal between the second housing section **108** and the second seal plate **202***a*. In some embodiments, the second static seal **204***a* is fabricated from an elastomer material. In the exemplary embodiment, the second static seal **204***a* is compressed between the second housing section **108** and the second seal plate **202***a*, so that the second static seal **204***a* provides a biasing force on the second seal plate **202***a* and the second static seal plate **200***a* in a direction toward the first louver plate **150***a*, which aids in maintaining a seal between the first louver plate **150***a* and the second housing section **108**.

[0071] In general, each of the second static seal plate **200***a*, the second seal plate **202***a*, and the second static seal **204***a* include similar features that are arranged in the same rotational orientation about the first central axis **156***a*, which aid in defining one or more of the sealed fluid channels or passageways between the first louver plate **150***a* and the second housing section **108**. Specifically, each of the second static seal plate **200***a*, the second seal plate **202***a*, and the second static seal **204***a* includes a second outer wall **206***a* and a second inner wall **208***a*. Each of the second static seal plate **200***a*, the second seal plate **202***a*, and the second static seal **204***a* define a generally annular shape that does not include includes any partitions, walls, or dividers that extend through the center of the annular shape (e.g., extends from the second inner wall **208***a* to divide the interior of the annular shape). As such, the second seal assembly **154***a* defines a single fluid passageway (e.g., a fourth fluid passageway **210***a* within the first louver plate assembly **112**). The fourth fluid passageway **210***a* extends between the first louver plate **150***a* and the fourth housing port **122**. In some embodiments, the second seal assembly **154***a* may define more than one fluid passageway (e.g., by including a particular number of internal partitions, walls, or dividers) to accommodate a particular porting configuration in the second housing section **108**.

[0072] In the exemplary embodiment, the second static seal plate **200***a* and the second seal plate **202***a* may include one more anti-rotation features that rotationally fix the second static seal plate **200***a* and the second seal plate **202***a* relative to the housing assembly **102** (i.e., prevent the second static seal plate **200***a* and the second seal plate **202***a* from rotating relative to the housing assembly **102**). For example, the second static seal plate **200***a* includes a lip **212***a* that protrudes outwardly from the second outer wall **206***a* of the second static seal plate **200***a*. In general, the lip **212***a* interrupts the round or circular shape of the second outer wall **206***a* of the second static seal plate **200***a*, and includes a pair of anti-rotation surfaces **214***a* arranged at opposing edges of the lip **212***a*. The second static seal plate **200***a* also includes a pair of recesses or notches **216***a* formed in the lip **212***a* between the anti-rotation surfaces **214***a*.

[0073] The second seal plate **202***a* includes a plurality of tabs **218***a* that extend outwardly from the second outer wall **206***a* of the second seal plate **202***a*. Each of the plurality of tabs **218***a* is received within a corresponding one of the notches **216***a* of the second static seal plate **200***a*, which aids in preventing relative rotation between the second static seal plate **200***a* and the second seal plate **202***a*. As described herein, the plurality of tabs **218***a* and/or the anti-rotation surfaces **214***a* may engage a structure within the housing assembly **102** to prevent relative rotation between the housing assembly **102** and both the second static seal plate **200***a* and the second seal plate **202***a*.

[0074] With reference to FIGS. **2** and **9-12**, in the exemplary embodiment, the second louver plate assembly **114** includes a second louver plate **150***b*, a first seal assembly **152***b* (e.g., an upper seal assembly from the perspective of FIG. **2**), and a second seal assembly **154***b* (e.g., a lower seal assembly from the perspective of FIG. **2**). The second louver plate assembly **114** is arranged along a second central axis **156***b*, which is defined along a central rod or pin **158***b*. The first or upper seal assembly **152***b* is arranged between the second louver plate **150***b* and the first housing section **106** (e.g., in a direction along the second central axis **156***b*), and the second seal assembly **154***b* is arranged between the first louver plate **150***a* and the second housing section **108**. In other words, the first seal assembly **152***b* and the second seal assembly **154***b* are arranged on opposing sides of the second louver plate **150***b*.

[0075] The second louver plate **150***b* defines a generally circular or round shape and includes a plurality of gear teeth arranged around an outer perimeter of the second louver plate **150***b*. A cutout, aperture, hole, slot, shown as louver cutout **160***b* extends through the second louver plate **150***b*. Specifically, the louver cutout **160***b* extends through both an upper or first louver surface **162***b* and a lower or second louver surface **164***b*. In the exemplary embodiment, the louver cutout

160*b* defines a generally triangular shape, with two straight edges and one convex or rounded edge. In some embodiments, the louver cutout **160***b* may be shaped differently (e.g., round, circular, rectangular, or another shape) to provide the appropriate connections between the sealed fluid channels or fluid passageways formed within the second louver plate assembly **114**

[0076] In general, the first seal assembly **152***b* is sealed between the second louver plate **150***b* and the first housing section **106** and defines one or more fluid channels or fluid passageways, each extending between the second louver plate **150***b* and one of the fifth housing port **124** and the sixth housing port **126**. With specific reference to FIGS. **2**, **9**, and **10**, in the exemplary embodiment, the first seal assembly **152***b* includes a first static seal plate **166***b*, a first seal plate **168***b*, and a first static seal **170***b*. When the second louver plate assembly **114** is assembled, the first static seal plate **166***b* is in engagement with and sealed against the upper or first louver surface **162***b* of the second louver plate **150***b* (see, e.g., FIG. **17**). The first seal plate **168***b* is arranged between the first static seal plate **166***b* and the first static seal **170***b* (e.g., in a direction along the second central axis **156***b*), and is sealed against both the first static seal plate **166***b* and the first static seal **170***b*.

[0077] The first static seal **170***b* forms a seal between the first housing section **106** and the first seal plate **168***b*. In some embodiments, the first static seal **170***b* is fabricated from an elastomer material. In the exemplary embodiment, the first static seal **170***b* is compressed between the first housing section **106** and the first seal plate **168***b*, so that the first static seal **170***b* provides a biasing force on the first seal plate **168***b* and the first static seal plate **166***b* in a direction toward the second louver plate **150***b*, which aids in maintaining a seal between the second louver plate **150***b* and the first housing section **106**.

[0078] In general, each of the first static seal plate **166***b*, the first seal plate **168***b*, and the first static seal **170***b* include similar features that are arranged in the same rotational orientation about the second central axis **156***b*, which aid in defining one or more of the sealed fluid channels or passageways between the second louver plate **150***b* and the first housing section **106**. Specifically, each of the first static seal plate **166***b*, the first seal plate **168***b*, and the first static seal **170***b* includes a first outer wall **172***b*, a first inner wall **174***b*, a first central hub **176***b*, and a plurality of partitions, walls, or dividers, shown as first inner walls **178***b*. Each of the first inner walls **178***b* extends radially (e.g., relative to the second central axis **156***b*) between the first inner wall **174***b* and the first central hub **176***b*.

[0079] In general, a number of the first inner walls **178***b* within the first seal assembly **152***b* defines the number of sealed fluid channels or passageways formed within the first seal assembly **152***b* (e.g., one or less of the inner walls defines one fluid passageway, two of the inner walls defines two fluid passageways, three of the inner walls defines three fluid passageways, etc.). In the exemplary embodiment, the first seal assembly **152***b* includes two of the first inner walls **178***b*, which divides the flow area between the second louver plate **150***b* and the first housing section **106** into a first fluid passageway **180***b* and a second fluid passageway **182***b*. The first fluid passageway **180***b* extends between the second louver plate **150***b* and the fifth housing port **124**, and the second fluid passageway **182***b* extends between the second louver plate **150***b* and the sixth housing port **126**. In some embodiments, the first seal assembly **152***b* may include more or less than two of the first inner walls **178***b* to define more or less than two fluid passageways to accommodate a particular porting configuration in the first housing section **106**.

[0080] In the exemplary embodiment, each of the first inner walls **178***b* is rotationally or circumferentially spaced from one another. The circumferential spacing between adjacent pairs of the first inner walls **178***b* defines circumferential length of each of the first fluid passageway **180***b* and the second fluid passageway **182***b* (e.g., a rotational angle about the second central axis **156***b* over which the louver cutout **160***b* may travel to allow fluid flow through the respective passageways formed in the first seal assembly **152***b*). In the exemplary embodiment, a circumferential length **186***b* of the first fluid passageway **180***b* may be different than a circumferential length **188***b* of the second fluid passageway **182***b*. In some embodiments, the circumferential length **186***b* of the first fluid passageway **180***b* may be approximately equal to the circumferential length **188***b* of the second fluid passageway **182***b*.

[0081] In the exemplary embodiment, the first static seal plate **166***b* and the first seal plate **168***b* may include one more anti-rotation features that rotationally fix the first static seal plate **166***b* and the first seal plate **168***b* relative to the housing assembly **102** (i.e., prevent the first static seal plate **166***b* and the first seal plate **168***b* from rotating relative to the housing assembly **102**). For example, the first static seal plate **166***b* includes a lip **192***b* that protrudes outwardly from the first outer wall **172***b* of the first static seal plate **166***b*. In general, the lip **192***b* interrupts the round or circular shape of the first outer wall **172***b* of the first static seal plate **166***b*, and includes a pair of anti-rotation surfaces **194***b* arranged at opposing edges of the lip **192***b*. The first static seal plate **166***b* also includes a pair of recesses or notches **196***b* formed in the lip **192***b* between the anti-rotation surfaces **194***b*.

[0082] The first seal plate **168***b* includes a plurality of tabs **198***b* that extend outwardly from the first outer wall **172***b* of the first seal plate **168***b*. Each of the notches **196***b* receives one of the plurality of tabs **198***b*, which aids in preventing relative rotation between the first static seal plate **166***b* and the first seal plate **168***b*. As described herein, the plurality of tabs **198***b* and/or the anti-rotation surfaces **194***b* may engage a structure within the housing assembly **102** to prevent relative rotation between the housing assembly **102** and both the first static seal plate **166***b* and the first seal plate **168***b*.

[0083] In general, the second seal assembly **154***b* is sealed between the second louver plate **150***b* and the second housing section **108** and defines one or more fluid channels or fluid passageways, each extending between the second louver plate **150***b* and one or more of the seventh housing port **128** and the eighth housing port **130**. With specific reference to FIGS. **2**, **11**, and **12**, in the exemplary embodiment, the second seal assembly **154***b* includes a second static seal plate **200***b*, a second seal plate **202***b*, and a second static seal **204***b*. When the second louver plate assembly **114** is assembled, the second static seal plate **200***b* is in engagement with and sealed against the lower or second louver surface **164***b* of the second louver plate **150***b* (see, e.g., FIG. **17**). The second seal plate **202***b* is arranged between the second static seal plate **200***b* and the second static seal **204***b* (e.g., in a direction along the second central axis **156***b*), and is sealed against both the second static seal plate **200***b* and the second static seal **204***b*.

[0084] The second static seal **204***b* forms a seal between the second housing section **108** and the second seal plate **202***b*. In some embodiments, the second static seal **204***b* is fabricated from an elastomer material. In the exemplary embodiment, the second static seal **204***b* is compressed between the second housing section **108** and the second seal plate **202***b*, so that the second static seal **204***b* provides a biasing force on the second seal plate **202***b* and the second static seal plate **200***b* in a direction toward the second louver plate **150***b*, which aids in maintaining a seal between the second louver plate **150***b* and the second housing section **108**.

[0085] In general, each of the second static seal plate **200***b*, the second seal plate **202***b*, and the second static seal **204***b* include similar features that are arranged in the same rotational orientation about the second central axis **156***b*, which aid in defining one or more of the sealed fluid channels or passageways between the second louver plate **150***b* and the second housing section **108**. Specifically, each of the second static seal plate **200***b*, the second seal plate **202***b*, and the second static seal **204***b* includes a second outer wall **206***b*, a second inner wall **208***b*, a second central hub **209***b*, and a plurality of partitions, walls, or dividers, shown as second inner walls **211***b*. Each of the second inner walls **211***b* extends radially (e.g., relative to the second central axis **156***b*) between the second inner wall **208***b* and the second central hub **209***b*.

[0086] In general, a number of the second inner walls **211***b* within the second seal assembly **154***b* defines the number of sealed fluid channels or passageways formed within the second seal assembly **154***b* (e.g., one or less of the inner walls defines one fluid passageway, two of the inner walls defines two fluid passageways, three of the inner walls defines three fluid passageways, etc.).

In the exemplary embodiment, the second seal assembly **154***b* includes two of the second inner walls **211***b*, which divides the flow area between the second louver plate **150***b* and the second housing section **108** into a third passageway **213***b* and a fourth fluid passageway **215***b*. The third passageway **213***b* extends between the second louver plate **150***b* and the seventh housing port **128**, and the fourth fluid passageway **215***b* extends between the second louver plate **150***b* and the eighth housing port **130**. In some embodiments, the second seal assembly **154***b* may include more or less than two of the second inner walls **211***b* to define more or less than two fluid passageways to accommodate a particular porting configuration in the second housing section **108**.

[0087] In the exemplary embodiment, each of the second inner walls **211***b* is rotationally or circumferentially spaced from one another. The circumferential spacing between adjacent pairs of the first inner walls second inner walls **211***b* defines circumferential length of each of the third passageway **213***b* and the fourth fluid passageway **215***b* (e.g., a rotational angle about the second central axis **156***b* over which the louver cutout **160***b* may travel to allow fluid flow through the respective passageways formed in the second seal assembly **154***b*). In the exemplary embodiment, a circumferential length **217***b* of the third passageway **213***b* may be different than a circumferential length **219***b* of the fourth fluid passageway **215***b*. In some embodiments, the circumferential length **217***b* of the third passageway **213***b* may be approximately equal to the circumferential length **219***b* of the fourth fluid passageway **215***b*.

[0088] In the exemplary embodiment, the first seal assembly **152***b* defines a different rotational orientation about the second central axis **156***b* than the second seal assembly **154***b* (see, e.g., FIG. **2**), which allows different porting connections between the first housing section **106** and the second housing section **108**. For example, the rotational orientation (e.g., circumferential position) of each of the first inner walls **178***b* is different that the rotational orientation (e.g., circumferential position) of each of the second inner walls **211***b*. In this way, for example, the second louver plate assembly **114** is able to provide or inhibit fluid communication between different porting connections and arrangements on the first housing section **106** and the second housing section **108**. Additionally, the two passageways formed in each of the first seal assembly **152***b* and the second seal assembly **154***b* may be partitioned, via the louver cutout **160***b*, to provide more than two fluid flow paths through the second louver plate assembly **114**. In some embodiments, the first seal assembly **152***b* may define approximately the same rotational orientation as the second seal assembly **154***b*, depending on the particular porting arrangement and thermal management application.

[0089] In the exemplary embodiment, the second static seal plate **200***b* and the second seal plate **202***b* may include one more anti-rotation features that rotationally fix the second static seal plate **200***b* and the second seal plate **202***b* relative to the housing assembly **102** (i.e., prevent the second static seal plate **200***b* and the second seal plate **202***b* from rotating relative to the housing assembly **102**). For example, the second static seal plate **200***b* includes a lip **212***b* that protrudes outwardly from the second outer wall **206***b* of the second static seal plate **200***b*. In general, the lip **212***b* interrupts the round or circular shape of the second outer wall **206***b* of the second static seal plate **200***b*, and includes a pair of anti-rotation surfaces **214***b* arranged at opposing edges of the lip **212***b*. The second static seal plate **200***b* also includes a recess or notch **216***b* formed in the lip **212***b* between the anti-rotation surfaces **214***b*.

[0090] The second seal plate **202***b* includes a tab **218***b* that extends outwardly from the second outer wall **206***b* of the second seal plate **202***b*. The tab **218***b* is received within the notch **216***b* of the second static seal plate **200***b*, which aids in preventing relative rotation between the second static seal plate **200***b* and the second seal plate **202***b*. As described herein, the tab **218***b* and/or the anti-rotation surfaces **214***b* may engage a structure within the housing assembly **102** to prevent relative rotation between the housing assembly **102** and both the second static seal plate **200***b* and the second seal plate **202***b*.

[0091] Turning to FIGS. **13-17**, the shape and number of passageways within first louver plate

assembly **112** and the second louver plate assembly **114** generally conforms to the number, shape, and arrangement of the ports within the housing assembly **102**. The first louver plate assembly **112** is sealed between the first housing section **106** and the first louver plate **150***a*, and between the second housing section **108** and the first louver plate **150***a*. As shown in FIG. **13**, the first housing section **106** includes a recess **220** for the first seal assembly **152***a* that is dimensioned to at least partially receive the first static seal **170***a* therein and a recess **222** for the first seal assembly **152***b* that is dimensioned to at least partially receive the first static seal **170***b* therein. In general, the internal walls that form the recess **220** conform to the shape and profile of the first static seal **170***b* so that the first static seal **170***b* may be pressed into the recess **220** during assembly of the thermal management control valve **100**. In some embodiments, the first static seal **170***b* may be a press-in-place seal. A central bore **224** is arranged at an axial center (e.g., along the first central axis **156***a*) of the recess **220** and extends axially in a direction toward the first louver plate **150***a*. When assembled, the central bore **224** receives an end of the pin **158***a* (see, e.g., FIG. **17**), which extends through an axial center of the first louver plate **150***a*.
[0092] The internal walls that form the recess **222** conform to the shape and profile of the first static seal **170***b* so that the first static seal **170***b* may be pressed into the recess **222** during assembly of the thermal management control valve **100**. In some embodiments, the first static seal **170***b* may be a press-in-place seal. A central bore **226** is arranged at an axial center (e.g., along the second central axis **156***b*) of the recess **222** and extends axially in a direction toward the second louver plate **150***b*. When assembled, the central bore **226** receives an end of the pin **158***b* (see, e.g., FIG. **17**), which extends through an axial center of the second louver plate **150***b*.
[0093] As shown in FIG. **14**, the second housing section **108** includes a recess **228** for the second seal assembly **154***a* that is dimensioned to at least partially receive the second static seal **204***a* therein and a recess **230** for the second seal assembly **154***b* that is dimensioned to at least partially receive the second static seal **204***b* therein. In general, the internal walls that form the recess **228** conform to the shape and profile of the second static seal **204***a* so that the second static seal **204***a* may be pressed into the recess **228** during assembly of the thermal management control valve **100**. In some embodiments, the second static seal **204***a* may be a press-in-place seal. In the exemplary embodiment, the recess **228** defines a generally annular shape, like the second static seal **204***a*, and a plurality of support arms **232** extend between an internal wall of the fourth housing port **122** and a central bore **234**. The central bore **234** is arranged at an axial center (e.g., along the first central axis **156***a*) of the fourth housing port **122**. The central bore **234** extends axially in a direction toward the first louver plate **150***a* and, when the thermal management control valve **100** is assembled, the central bore **234** receives an end of the pin **158***a* (see, e.g., FIG. **17**).
[0094] The internal walls that form the recess **230** conform to the shape and profile of the second static seal **204***b* so that the second static seal **204***b* may be pressed into the recess **230** during assembly of the thermal management control valve **100**. In some embodiments, the second static seal **204***b* may be a press-in-place seal. A central bore **236** is arranged at an axial center (e.g., along the second central axis **156***b*) of the recess **230** and extends axially in a direction toward the second louver plate **150***b*. When assembled, the central bore **236** receives an end of the pin **158***b* (see, e.g., FIG. **17**).
[0095] With reference to FIGS. **14-16**, the second housing section **108** includes one or more features that aid in preventing the first seal assembly **152***a*, the first seal assembly **152***b*, the second seal assembly **154***a*, and the second seal assembly **154***b* from rotating relative to the housing assembly **102** (and specifically relative to the second housing section **108**). Specifically, the second housing section **108** includes a plurality of first tab recesses **238**, a plurality of first stop surfaces **240**, a plurality of second tab recesses **242**, and a plurality of second stop surface **244**. The plurality of first tab recesses **238** are each recessed into an internal wall **246** of the second housing section **108**. The plurality of first tab recesses **238** are rotationally spaced from one another (e.g., about the first central axis **156***a*) along the internal wall **246**. When the thermal management control valve

100 is assembled, each of the plurality of first tab recesses **238** receives a corresponding one of the plurality of tabs **198***a* of the first seal plate **168***a* (see, e.g., FIG. **15**). The interface between the plurality of first tab recesses **238** and the plurality of tabs **198***a* aids in limiting or preventing relative rotation between the first seal plate **168***a* and the second housing section **108**. In the exemplary embodiment, two of the plurality of tabs **198***a* are received within the plurality of first tab recesses **238**, and one of the plurality of tabs **198***a* is at least partially received within the notch **196***a* of the first static seal plate **166***a*. In some embodiments, the first seal plate **168***a* may include more or less than three of the plurality of tabs **198***a* arranged in varying orientations, and a corresponding number of both the notches **196***a* in the first static seal plate **166***a* and the plurality of first tab recesses **238** in the second housing section **108**.

[0096] The plurality of first stop surfaces **240** are each arranged within the louver manifold **140** and are generally opposed to one another. In this way, for example, the plurality of first stop surfaces **240** provide rotational stops for both rotational directions that the first louver plate **150***a* may be rotated. When the thermal management control valve **100** is assembled, each of the anti-rotation surfaces **194***a* of the first static seal plate **166***a* abuts a corresponding one of the plurality of first stop surfaces **240** (see, e.g., FIG. **15**), which limits or prevents relative rotation between the first static seal plate **166***a* and the second housing section **108**. Accordingly, the first static seal plate **166***a* and the first seal plate **168***a* both include anti-rotation features that interface with the second housing section **108**, and with one another, to prevent or limit rotation relative to the second housing section **108**.

[0097] For the second seal assembly **154***a*, each of the anti-rotation surfaces **214***a* of the second static seal plate **200***a* abuts a corresponding one of the plurality of first stop surfaces **240** (see, e.g., FIG. **16**), which limits or prevents relative rotation between the second static seal plate **200***a* and the second housing section **108**. In the exemplary embodiment, the plurality of tabs **218***a* of the second seal plate **202***a* are at least partially received within the notches **216***a* formed in the second static seal plate **200***a*, which limits or prevents relative rotation between the second static seal plate **200***a* and the second seal plate **202***a* and, thereby, between the second seal plate **202***a* and the second housing section **108**. In some embodiments, the outer edges or surfaces of the plurality of tabs **218***a* also abut a corresponding one of the plurality of first stop surfaces **240** to further limit or prevent relative rotation between the second seal plate **202***a* and the second housing section **108**. Accordingly, the second static seal plate **200***a* and the second seal plate **202***a* both include anti-rotation features that interface with the second housing section **108**, and with one another, to prevent or limit rotation relative to the second housing section **108**.

[0098] For the first seal assembly **152***b*, the plurality of second tab recesses **242** are each recessed into an internal wall **248** of the second housing section **108**. The plurality of second tab recesses **242** are rotationally spaced from one another (e.g., about the second central axis **156***b*) along the internal wall **248**. When the thermal management control valve **100** is assembled, each of the plurality of second tab recesses **242** receives a corresponding one of the plurality of tabs **198***b* of the first seal plate **168***b* (see, e.g., FIG. **15**). The interface between the plurality of second tab recesses **242** and the plurality of tabs **198***b* aids in limiting or preventing relative rotation between the first seal plate **168***b* and the second housing section **108**. In the exemplary embodiment, two of the plurality of tabs **198***b* are received within the plurality of second tab recesses **242**, and two of the plurality of tabs **198***b* are least partially received within the notches **196***b* of the first static seal plate **166***b*. In some embodiments, the first seal plate **168***b* may include more or less and four of the plurality of tabs **198***b* arranged in varying orientations, and a corresponding number of both the notches **196***b* in the first static seal plate **166***b* and the plurality of second tab recesses **242** in the second housing section **108**.

[0099] The plurality of second stop surface **244** are each arranged within the louver manifold **140** and are generally opposed to one another. In this way, for example, the plurality of second stop surface **244** provide rotational stops for both rotational directions that the first louver plate **150***a* may be rotated. When the thermal management control valve **100** is assembled, each of the anti-rotation surfaces **194***b* of the first static seal plate **166***b* abuts a corresponding one of the plurality of second stop surface **244** (see, e.g., FIG. **15**), which limits or prevents relative rotation between the first static seal plate **166***b* and the second housing section **108**. Accordingly, the first static seal plate **166***b* and the first seal plate **168***b* both include anti-rotation features that interface with the second housing section **108**, and with one another, to prevent or limit rotation relative to the second housing section **108**.

[0100] For the second seal assembly **154***b*, each of the anti-rotation surfaces **214***b* of the second static seal plate **200***b* abuts a corresponding one of the plurality of second stop surface **244** (see, e.g., FIG. **16**), which limits or prevents relative rotation between the second static seal plate **200***b* and the second housing section **108**. In the exemplary embodiment, the tab **218***b* of the second seal plate **202***a* is at least partially received within the notch **216***b* formed in the second static seal plate **200***b*, which limits or prevents relative rotation between the second static seal plate **200***b* and the second seal plate **202***b* and, thereby, between the second seal plate **202***b* and the second housing section **108**. Accordingly, the second static seal plate **200***b* and the second seal plate **202***b* both include anti-rotation features that interface with the second housing section **108**, and with one another, to prevent or limit rotation relative to the second housing section **108**.

[0101] With reference to FIGS. **2** and **15-17**, in the exemplary embodiment, the drive shaft **134** includes a threaded portion **250** (e.g., helical external threads) that are threaded with gear teeth formed on the outer periphery of both the first louver plate **150***a* and the second louver plate **150***b* (see, e.g., FIG. **16**). In this way, for example, selective rotation of the drive shaft **134**, via the rotary actuator **132**, results in rotation of the first louver plate **150***a* about the first central axis **156***a* in a first rotational direction **252**, and rotation of the second louver plate **150***b* about the second central axis **156***b* in a second rotational direction **254** opposite to the first rotational direction. The selective rotation of the first louver plate **150***a* and the second louver plate **150***b* results in rotation of the louver cutout **160***a* and the louver cutout **160***b*, respectively, relative to the various fluid passageways formed by the first louver plate assembly **112** and the second louver plate assembly **114**. Accordingly, the louver cutout **160***a* and the louver cutout **160***b* are selectively rotated to various rotational positions to provide or inhibit fluid communication along the various fluid passageways formed in the first louver plate assembly **112** and the second louver plate assembly **114**, and thereby provide or inhibit fluid communication between the ports formed in the housing assembly **102**. For example, in the exemplary position of FIG. **17**, the first louver plate **150***a* blocks fluid communication between the first housing port **116** and the fourth housing port **122** (e.g., fluid flow is blocked along the first fluid passageway **180***a* and the fourth fluid passageway **210***a*), and the second louver plate **150***b* provides fluid communication between the fifth housing port **124** and the eighth housing port **130** (e.g., fluid flow is allowed along the first fluid passageway **180***b* and the fourth fluid passageway **215***b*). It should be appreciated that the worm drive-like actuation or movement of the first louver plate **150***a* and the second louver plate **150***b* driven by the threaded interaction between the drive shaft **134** and both the first louver plate **150***a* and the second louver plate **150***b* is but one exemplary embodiment for actuating or rotating the first louver plate **150***a* and the second louver plate **150***b*. In some embodiments, the rotary actuator **132** may rotate a spur gear, a cam, or an equivalent mechanism that converts rotational motion to rotational motion to selectively rotate the first louver plate **150***a* and the second louver plate **150***b*.

[0102] In general, the design and properties of the thermal management control valve **100** including the first louver plate assembly **112** and the second louver plate assembly **114** enables fluid communication to be provided or inhibited between various combination of porting configurations. In addition, the partitioning of the fluid passageways within the first louver plate assembly **112** and the second louver plate assembly **114** and the selective rotation of the first louver plate **150***a* and second louver plate **150***b* enables the connections between various ports to be proportionally/incrementally opened and/or closed (e.g., the flow area of a particular flow path proportionally/incrementally increases or decreases). As described herein, the ports of the housing assembly **102** may each be in fluid communication with a connector, fitting, or tubing that delivers fluid flow to or from a thermal management component. FIGS. **18-20** show an exemplary embodiment of the thermal management control valve **100** with the housing assembly **102** including a plurality of connectors or fittings, shown as a plurality of ports, each being in fluid communication with one or more of the first housing port **116**, the second housing port **118**, the third housing port **120**, the fourth housing port **122**, the fifth housing port **124**, the sixth housing port **126**, the seventh housing port **128**, and the eighth housing port **130**. Specifically, the housing assembly **102** includes a first port **256**, a second port **258**, a third port **260**, a fourth port **262**, a fifth port **264**, and a sixth port **266**. In the exemplary embodiment, the first port **256**, the second port **258**, and the third port **260** are arranged on the first housing section **106**, and the fourth port **262**, the fifth port **264**, and the sixth port **266** are arranged on the second housing section **108**. As shown in FIG. **19**, the first port **256** is in fluid communication with the first housing port **116** (and the first fluid passageway **180***a*), the second port **258** is in fluid communication with both the second housing port **118** and the fifth housing port **124** (e.g., both the second fluid passageway **182***a* and the first fluid passageway **180***b*), and the third port **260** is in fluid communication with both the third housing port **120** and the sixth housing port **126** (e.g., both the third fluid passageway **184***a* and the second fluid passageway **182***b*). As shown in FIG. **20**, the fourth port **262** is in fluid communication with the fourth housing port **122** (and the fourth fluid passageway **210***a*), the fifth port **264** is in fluid communication with the seventh housing port **128** (and the third passageway **213***b*), and the sixth port **266** is in fluid communication with the eighth housing port **130** (and the fourth fluid passageway **215***b*). It should be appreciated that the porting of FIGS. **18-20** (e.g., including six of the ports, with two of the ports being connected to two of the housing ports and the remaining ports each being connected to one of the housing ports) is but one exemplary porting configuration and the housing assembly **102** may be adapted to accommodate alternative porting configurations as needed for particular thermal management application. For example, each of the housing ports may be in fluid communication with a corresponding one fluid port (e.g., the housing assembly **102** may include eight fluid ports). In some embodiments, the housing assembly **102** may include more or less that six fluid ports.

[0103] In general, during operation, the first louver plate **150***a* and the second louver plate **150***b* are selectively rotatable between a plurality of rotational positions to provide or inhibit fluid communication between one or more of the ports of the housing assembly **102**. For example, turning to FIGS. **21-28**, the first louver plate assembly **112** and the second louver plate assembly **114** are shown with the first louver plate **150***a* and the second louver plate **150***b* in a plurality of rotational positions, each including a particular set of fluid flow paths between the various ports of the housing assembly **102**. In FIGS. **21-28**, each of the views is from a top of the thermal management control valve **100** and the features of second housing section **108**, the second seal assembly **154***a*, and the second seal assembly **154***b* (e.g., features below the first louver plate **150***a* and the second louver plate **150***b*) are shown in dashed lines. While FIGS. **21-28** illustrate the first louver plate **150***a* and the second louver plate **150***b* in specific positions, it should be appreciated that the rotary actuator **132** may selectively move the first louver plate **150***a* and the second louver plate **150***b* to any rotary position, including any rotary position between the positions illustrated in FIGS. **21-28**.

[0104] FIG. **21** shows the flow control assembly **104** in a first operating mode (e.g., with the first louver plate **150***a* and the second louver plate **150***b* in a first position). In the first operating mode, the louver cutout **160***a* rotationally aligns with (e.g., allows/opens fluid flow through) the second housing port **118** (and the second fluid passageway **182***a*) and the fourth housing port **122** (and the fourth fluid passageway **210***a*), and fluid communication is provided between the second housing port **118** and the fourth housing port **122** (and between the second port **258** and the fourth port **262**). The first louver plate **150***a* blocks or closes the first housing port **116** and the third housing port **120**. In the first operating mode, the louver cutout **160***b* rotationally aligns with the sixth housing port **126** (e.g., and the second fluid passageway **182***b*) and the seventh housing port **128** (and the third passageway **213***b*), and fluid communication is provided between the sixth housing port **126** and the seventh housing port **128** (and between the third port **260** and the fifth port **264**). The second louver plate **150***b* blocks or closes the fifth housing port **124** and the eighth housing port **130**.

[0105] FIG. **22** shows the flow control assembly **104** in a first intermediate operating mode (e.g., between the first operating mode and a second operating mode, with the first louver plate **150***a* and the second louver plate **150***b* in a first intermediate position). In the first intermediate operating mode, the louver cutout **160***a* rotationally aligns with the second housing port **118** (and the second fluid passageway **182***a* and the fourth housing port **122** (and the fourth fluid passageway **210***a*), and fluid communication is provided between the second housing port **118** and the fourth housing port **122** (and between the second port **258** and the fourth port **262**). The first louver plate **150***a* blocks or closes the first housing port **116** and the third housing port **120**. In the first intermediate position, the louver cutout **160***b* rotationally aligns with the sixth housing port **126** (and the second fluid passageway **182***b*) and both the seventh housing port **128** (and the third passageway **213***b*) and the eighth housing port **130** (and the fourth fluid passageway **215***b*). Fluid communication is provided between the sixth housing port **126** and the seventh housing port **128** (and between the third port **260** and the fifth port **264**), and between the sixth housing port **126** and the eighth housing port **130** (and between the third port **260** and the sixth port **266**). With the louver cutout **160***b* overlapping with both the seventh housing port **128** and the eighth housing port **130**, the flow area is reduced when compared to these ports being fully opened (e.g., any entirety of the louver cutout **160***b* overlaps with the port) and partial fluid flow (e.g., less than a maximum fluid flow rate) is provided through the seventh housing port **128** and the eighth housing port **130**. In the first intermediate position, the second louver plate **150***b* blocks or closes the fifth housing port **124**.

[0106] FIG. **23** shows the flow control assembly **104** in a second operating mode (e.g., with the first louver plate **150***a* and the second louver plate **150***b* in a second position). In the second operating mode, the louver cutout **160***a* rotationally aligns with the second housing port **118** (and the second fluid passageway **182***a* and the fourth housing port **122** (and the fourth fluid passageway **210***a*), and fluid communication is provided between the second housing port **118** and the fourth housing port **122** (and between the second port **258** and the fourth port **262**). The first louver plate **150***a* blocks or closes the first housing port **116** and the third housing port **120**. In the first intermediate position, the louver cutout **160***b* rotationally aligns with the sixth housing port **126** (and the second fluid passageway **182***b*) and the eighth housing port **130** (and the fourth fluid passageway **215***b*), and fluid communication is provided between the sixth housing port **126** and the eighth housing port **130** (and between the third port **260** and the sixth port **266**). The second louver plate **150***b* blocks or closes the fifth housing port **124** and the seventh housing port **128** (and the third passageway **213***b*).

[0107] FIG. **24** shows the in a third operating mode (e.g., with the first louver plate **150***a* and the second louver plate **150***b* in a third position). In the third operating mode, the louver cutout **160***a* rotationally aligns with the third housing port **120** (and the third fluid passageway **184***a*) and the fourth housing port **122** (and the fourth fluid passageway **210***a*), and fluid communication is provided between the third housing port **120** and the fourth housing port **122** (and between the third port **260** and the fourth port **262**). The first louver plate **150***a* blocks or closes the first housing port **116** and the second housing port **118**. In the third operating mode, the louver cutout **160***b* rotationally aligns with the fifth housing port **124** (and the first fluid passageway **180***b*) and the eighth housing port **130** (and the fourth fluid passageway **215***b*), and fluid communication is provided between the fifth housing port **124** and the eighth housing port **130** (and between the second port **258** and the sixth port **266**). The second louver plate **150***b* blocks or closes the sixth housing port **126** and the seventh housing port **128**.

[0108] In an exemplary embodiment, the flow control assembly **104** is operable in a second intermediate position (e.g., between the second operating mode and the third operating mode, with the first louver plate **150***a* and the second louver plate **150***b* in a second intermediate position). In the second intermediate operating mode, the louver cutout **160***a* may rotationally align with both the second housing port **118** (and the second fluid passageway **182***a*) and the third housing port **120** (and the third fluid passageway **184***a*), and the fourth housing port **122** (and the fourth fluid passageway **210***a*). Fluid communication (e.g., partial fluid flow) is provided between the second housing port **118** and the fourth housing port **122** (and between the second port **258** and the fourth port **262**), and fluid communication (e.g., partial fluid flow) is provided between the third housing port **120** and the fourth housing port **122** (and between the third port **260** and the fourth port **262**). The first louver plate **150***a* may block or close the first housing port **116**. In the second intermediate position, the louver cutout **160***b* may rotationally align with both the fifth housing port **124** (and the first fluid passageway **180***b*) and the sixth housing port **126** (and the second fluid passageway **182***b*), and the eighth housing port **130** (and the fourth fluid passageway **215***b*). Fluid communication (e.g., partial fluid flow) is provided between the fifth housing port **124** and the eighth housing port **130** (and between the second port **258** and the sixth port **266**), and fluid communication (e.g., partial fluid flow) is provided between the sixth housing port **126** and the eighth housing port **130** (and between the third port **260** and the sixth port **266**). The second louver plate **150***b* may block or close the seventh housing port **128**.

[0109] FIG. **25** shows the flow control assembly **104** is operable in a third intermediate position (e.g., between the third operating mode and a fourth operating mode, with the first louver plate **150***a* and the second louver plate **150***b* in a third intermediate position). In the third intermediate operating mode, the louver cutout **160***a* rotationally aligns with the third housing port **120** (and the third fluid passageway **184***a*) and the fourth housing port **122** (and the fourth fluid passageway **210***a*), and fluid communication is provided between the third housing port **120** and the fourth housing port **122** (and between the third port **260** and the fourth port **262**). The first louver plate **150***a* blocks or closes the first housing port **116** and the second housing port **118**. In the third intermediate position, the louver cutout **160***b* rotationally aligns with the fifth housing port **124** (and the first fluid passageway **180***b*) and both the seventh housing port **128** (and the third passageway **213***b*) and the eighth housing port **130** (and the fourth fluid passageway **215***b*). Fluid communication (e.g., partial fluid flow) is provided between the fifth housing port **124** and the seventh housing port **128** (and between the second port **258** and the fifth port **264**), and fluid communication (e.g., partial fluid flow) is provided between the fifth housing port **124** and the eighth housing port **130** (and between the second port **258** and the sixth port **266**). The louver cutout **160***b* blocks or closes the sixth housing port **126**.

[0110] FIG. **26** shows the flow control assembly **104** in a fourth operating mode (e.g., with the first louver plate **150***a* and the second louver plate **150***b* in a fourth position). In the fourth operating mode, the louver cutout **160***a* rotationally aligns with the third housing port **120** (and the third fluid passageway **184***a*) and the fourth housing port **122** (and the fourth fluid passageway **210***a*), and fluid communication is provided between the third housing port **120** and the fourth housing port **122** (and between the third port **260** and the fourth port **262**). The first louver plate **150***a* blocks or closes the first housing port **116** and the second housing port **118**. In the fourth position, the louver cutout **160***b* rotationally aligns the fifth housing port **124** (and the first fluid passageway **180***b*) and the seventh housing port **128** (and the third passageway **213***b*), and fluid communication is provided between the fifth housing port **124** and the seventh housing port **128** (and between the second port **258** and the fifth port **264**). The second louver plate **150***b* blocks or closes the sixth housing port **126** and the eighth housing port **130**.

[0111] FIG. **27** shows the flow control assembly **104** in a fourth intermediate position (e.g., between the fourth operating mode and fifth fourth operating mode, with the first louver plate **150***a* and the second louver plate **150***b* in a fourth intermediate position). In the fourth intermediate operating mode, the louver cutout **160***a* rotationally aligns with both the first housing port **116** (and the first fluid passageway **180***a*) and the third housing port **120** (and the third fluid passageway **184***a*), and the fourth housing port **122**. Fluid communication (e.g., partial fluid flow) is provided between the first housing port **116** and the fourth housing port **122** (and between the first port **256** and the fourth port **262**), and fluid communication (e.g., partial fluid flow) is provided between the third housing port **120** and the fourth housing port **122** (and between the third port **260** and the fourth port **262**). The first louver plate **150***a* blocks or closes the second housing port **118**. In the fourth intermediate position, the louver cutout **160***b* rotationally aligns with the fifth housing port **124** (and the first fluid passageway **180***b*) and the seventh housing port **128** (and the third passageway **213***b*), and fluid communication is provided between the fifth housing port **124** and the seventh housing port **128** (and between the second port **258** and the fifth port **264**). The second louver plate **150***b* blocks or closes the sixth housing port **126** and the eighth housing port **130**.

[0112] FIG. **28** shows the flow control assembly **104** in a fifth operating mode (e.g., with the first louver plate **150***a* and the second louver plate **150***b* in a fifth position). In the fifth operating mode, the louver cutout **160***a* rotationally aligns with the first housing port **116** (and the first fluid passageway **180***a*) and the fourth housing port **122** (and the fourth fluid passageway **210***a*), and fluid communication is provided between the first housing port **116** and the fourth housing port **122** (and between the first port **256** and the fourth port **262**). The first louver plate **150***a* blocks or closes the second housing port **118** and the third housing port **120**. In the fifth position, the louver cutout **160***b* rotationally aligns with the fifth housing port **124** (and the first fluid passageway **180***b*) and the seventh housing port **128** (and the third passageway **213***b*), and fluid communication is provided between the fifth housing port **124** and the seventh housing port **128** (and between the second port **258** and the fifth port **264**). The second louver plate **150***b* blocks or closes the sixth housing port **126** and the eighth housing port **130**.

[0113] As described above, the design and properties of the first louver plate assembly **112** and the second louver plate assembly **114** enable the thermal management control valve **100** to provide various fluid connections between the ports of the housing assembly **102**. In some operating modes, both the first louver plate assembly **112** and the second louver plate assembly **114** provide fluid communication between two different ports. In other operating modes, at least one of the first louver plate assembly **112** and the second louver plate assembly **114** provides fluid communication between three different ports. In general, each of the operating modes (or positions of the first louver plate **150***a* and the second louver plate **150***b*) defines a unique set of connections between four or more of the housing ports (e.g., between the first housing port **116**, the second housing port **118**, the third housing port **120**, the fourth housing port **122**, the fifth housing port **124**, the sixth housing port **126**, the seventh housing port **128**, and the eighth housing port **130**) or between four or more of the fluid ports (e.g., the first port **256**, the second port **258**, the third port **260**, the fourth port **262**, the fifth port **264**, and the sixth port **266**). Each individual louver assembly (e.g., the first louver plate assembly **112** or the second louver plate assembly **114**) selectively inhibits or provides fluid communication between at least four of the housing ports, depending on the rotations position of the first louver plate **150***a* or the second louver plate **150***b*. These operational characteristics of the thermal management control valve **100** enable the thermal management control valve **100** to control fluid flow to various thermal management components using a single valve with a smaller packaging size (e.g., when compared to conventional thermal control valves or the use of multiple valves).

[0114] As described herein, the rotary louver concept employed by the first louver plate assembly **112** and the second louver plate assembly **114** may be used to enable proportional/incremental flow control (e.g., partial fluid flow that increases or decreases in flow area, either from a closed state to an open state or visa versa). For example, as the first louver plate **150***a* rotates from the fourth position to the fourth intermediate position, the third housing port **120** transitions from a fully open state and proportionally closes off, while the first housing port **116** opens up. It should be appreciated that the rotary actuator **132** may rotate the first louver plate **150***a* to any number of positions between the fourth position and the fourth intermediate position to adjust the flow areas defined by the third housing port **120** and the first housing port **116** as desired. Similar proportional control is exemplified with the second louver plate **150***b* transitioning from the first position to the first intermediate position, and from the third position to the third intermediate position.

[0115] FIGS. **29-32** show an exemplary embodiment of a thermal control valve **300**, according to an exemplary embodiment. In general, the thermal control valve **300** is configured to proportionally open or close one or more ports to vary the flow area between two or more ports in a housing. The thermal control valve **300** includes a rotary actuator **302** (e.g., a rotary motor, a stepper motor, an electric motor, etc.) that is coupled to a drive shaft **304**, a louver plate **306**, a static seal plate **308**, and a static seal **310**. The drive shaft **304** is coupled to the louver plate **306** and defines a drive axis **312**. In the exemplary embodiment, the drive shaft **304** is formed integrally with (e.g., as a unitary component) the louver plate **306**. In some embodiments, the drive shaft **304** and the louver plate **306** are separate components that are coupled together.

[0116] In the exemplary embodiment, the louver plate **306** defines a generally circular or round shape and includes a cutout, aperture, or channel, shown as louver slot **314**. The louver slot **314** extends completely through the louver plate **306** (e.g., from a top surface to an opposing bottom surface) so that fluid flow can be provided through the louver slot **314**. In the exemplary embodiment, the louver slot **314** defines a generally arcuate or curved shape that extends circumferentially about the drive axis **312** approximately one hundred and eighty degrees. In some embodiments, the louver slot **314** may extend circumferentially more or less than one hundred and eighty degrees.

[0117] The static seal plate **308** is arranged axially between the louver plate **306** and the static seal **310**. In the exemplary embodiment, the static seal plate **308** is in engagement with a top or first surface **316** of the louver plate **306**, so that a seal is formed between the static seal plate **308** and the top or first surface **316** of the louver plate **306**. In the exemplary embodiment, the static seal plate **308** includes a plurality of cutouts, slots, or channels that each define a varying flow area in a circumferential direction about the drive axis **312**. Specifically, the static seal plate **308** includes a first cutout, aperture, or channel, shown as first slot **318** and a second cutout, aperture, or channel, shown as second slot **320**. The first slot **318** is rotationally symmetric to the second slot **320** about the drive axis **312**. In the exemplary embodiment, the first slot **318** and the second slot **320** each define a generally arrow-shaped slot that includes a tail or first portion **322** and a head or second portion **324**. The tail or first portion **322** defines a gradually increasing flow area as the first slot **318** extends circumferentially in a direction from the tail or first portion **322** toward the head or second portion **324**. Continuing circumferentially in a direction from the tail or first portion **322** toward the head or second portion **324**, the first slot **318** and the second slot **320** define a step-change increase in flow area at the interface between the tail or first portion **322** and the head or second portion **324**, where a radial width of the first slot **318** and the second slot **320** defines a step-change increase. Alternatively, moving circumferentially along the first slot **318** and the second slot **320** in a direction from the head or second portion **324** toward the tail or first portion **322**, the first slot **318** and the second slot **320** define a step-change decrease in flow area at the interface between the tail or first portion **322** and the head or second portion **324**. Accordingly, the flow area through the static seal plate **308** may be proportionally adjusted based on a rotational position of the louver plate **306** and, specifically, what portions of the first slot **318** and/or the second slot **320** are rotationally aligned with the louver slot **314**. In some embodiments, the first slot **318** and the second slot **320** may be shaped differently to define a different change in flow area as the louver plate **306** is rotated relative to the first slot **318** and the second slot **320**.

[0118] The static seal plate **308** is sealed between both the louver plate **306** and the static seal **310**. In the exemplary embodiment, the static seal plate **308** includes a first channel **326** and a second channel **328**. When assembled, the first channel **326** surrounds or borders the first slot **318** and the second channel **328** surrounds or borders the second slot **320**, so that both the first slot **318** and the second slot **320** are sealed between the louver plate **306** and a housing that the thermal control valve **300** is mounted within (see, e.g., FIG. **37**). In the exemplary embodiment, a spring **330** is received on the drive shaft **304** and is biased between a pair of spring flanges **332**. When assembled, the spring **330** is biased between the spring flanges **332** (and against a housing) so that the louver plate **306** applies a compressive force on the static seal plate **308** and the static seal **310**, which aids in maintaining a seal between each of the louver plate **306**, the static seal plate **308**, the static seal **310**, and the housing (see, e.g., FIG. **37**).

[0119] With reference to FIGS. **32-36**, the thermal control valve **300** may be mounted within a housing assembly **334** that includes a first chamber **336** having a first port **338**, a second chamber **340** having a second port **342**, and a third chamber **344**. The thermal control valve **300** is configured to selectively provide or inhibit fluid communication between the first chamber **336** and the third chamber **344** through the first port **338**, and selectively provide or inhibit fluid communication between the second chamber **340** and the third chamber **344** through the second port **342**. Additionally, as the louver plate **306** is selectively rotated relative to the first slot **318** and the second slot **320**, the flow area provided through the first port **338** and/or the second port **342** is proportionally controlled, depending on the rotational position of the louver slot **314** along the first slot **318** and/or the second slot **320**. For example, the louver plate **306** is selectively rotatable between a first position (FIG. **33**), a second position (FIG. **36**), and any position between the first position and the second position. In the first position, the louver slot **314** rotationally aligns with an entirety of the first slot **318**, so that fluid flow is provided through the first port **338** with a maximum flow area provided by the first slot **318** (e.g., the louver plate **306** does not block any portion of the first slot **318**). In the first position, the louver plate **306** blocks or closes the second slot **320**, which prevents fluid flow through the second port **342**.

[0120] As the louver plate **306** is rotated from the first position in a direction toward the second position (see, e.g., FIG. **34**), the louver plate **306** begins to cover or close the tail or first portion **322** of the first slot **318**, which decreases the flow area through the first port **338**. The louver slot **314** also begins to rotationally align with the tail or first portion **322** of the second slot **320**, which provides fluid flow through the second port **342** and gradually increases the flow area (e.g., from no flow) through the second port **342**. As the louver plate **306** continues to rotate toward the second position (see, e.g., FIG. **35**), the flow area through the first port **338** proportionally decreases as the louver plate **306** blocks or closes more of the first slot **318**, and the flow area through the second port **342** proportionally increases as the louver slot **314** rotationally aligns with more of the second slot **320**. Eventually, once the louver plate **306** reaches the second position, the louver plate **306** blocks or closes the first slot **318**, which prevents fluid flow through the first port **338**, and the louver slot **314** rotationally aligns with an entirety of the tail or first portion **322**, so that fluid flow is provided through the second port **342** with a maximum flow area provided by the tail or first portion **322** (e.g., the louver plate **306** does not block any portion of the tail or first portion **322**). In some embodiments, the louver plate **306** may rotate back and forth between the first position and the second position (e.g., from the first position to the second position, and then from the second position to the first position). In this way, for example, the flow area changes provided during the transition from the first position to the second position may be reversed (e.g., the tail or first portion **322** proportionally decreases in flow area, and the first slot **318** proportionally increases in flow area).

[0121] As utilized herein with respect to numerical ranges, the terms “approximately,” “about,” “substantially,” and similar terms generally mean +/−10% of the disclosed values. When the terms “approximately,” “about,” “substantially,” and similar terms are applied to a structural feature (e.g., to describe its shape, size, orientation, direction, etc.), these terms are meant to cover minor variations in structure that may result from, for example, the manufacturing or assembly process and are intended to have a broad meaning in harmony with the common and accepted usage by

those of ordinary skill in the art to which the subject matter of this disclosure pertains. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.
[0122] It should be noted that the term "exemplary" and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).
[0123] The term "coupled" and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If "coupled" or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of "coupled" provided above is modified by the plain language meaning of the additional term (e.g., "directly coupled" means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of "coupled" provided above. Such coupling may be mechanical, electrical, or fluidic.
[0124] References herein to the positions of elements (e.g., "top," "bottom," "above," "below") are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.
[0125] The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single-or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.
[0126] The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program

products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0127] Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

[0128] It is important to note that the construction and arrangement of the thermal management control valves **100**, **200** as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein.

Claims

1. A thermal management control valve comprising: a housing assembly including a first housing section and a second housing section; a rotary actuator coupled to a drive shaft that extends into the housing assembly; and a louver plate assembly enclosed within the housing assembly and including: a louver plate including a louver cutout extending through the louver plate, wherein the louver plate is coupled to the drive shaft so that selective rotation of the drive shaft is configured to rotate the louver plate; a first seal assembly arranged between the louver plate and the first housing section, wherein the first seal assembly includes a plurality of inner walls that define a plurality of first fluid passageways between the louver plate and the first housing section; and a second seal assembly arranged between the louver plate and the second housing section, wherein the second seal assembly defines a second fluid passageway between the louver plate and the second housing section, wherein selective rotation of the louver plate rotationally aligns the louver cutout with one or more of the plurality of first fluid passageways and the second fluid passageway to provide fluid communication between the one or more of the plurality of first fluid passageways and the second fluid passageway.

2. The thermal management control valve of claim 1, wherein the plurality of first fluid passageways includes two or more fluid passageways.

3. The thermal management control valve of claim 1, wherein the first seal assembly includes: a first static seal plate in engagement with a first surface of the louver plate; a first seal plate; and a first static seal sealed between the first housing section and the first seal plate, wherein the first seal plate is arranged between the first static seal plate and the first static seal.

4. The thermal management control valve of claim 3, wherein the plurality of inner walls are formed by each of the first static seal plate, the first seal plate, and the first static seal plate.

5. The thermal management control valve of claim 3, wherein the first static seal plate includes an

anti-rotation surface that abuts a corresponding stop surface within the housing assembly to limit rotation of the first static seal plate relative to the housing assembly.
6. The thermal management control valve of claim 3, wherein the first seal plate includes a plurality of tabs that protrude outwardly from an outer wall of the first seal plate, and wherein each of the plurality of tabs is received within a corresponding recess formed in the housing assembly.
7. The thermal management control valve of claim 3, wherein the second seal assembly includes: a second static seal plate in engagement with a second surface of the louver plate; a second seal plate; and a second static seal sealed between the second housing section and the second seal plate, wherein the second seal plate is arranged between the second static seal plate and the second static seal.
8. The thermal management control valve of claim 7, wherein each of the second static seal plate, the second seal plate, and the second static seal defines an annular shape.
9. The thermal management control valve of claim 7, wherein each of the second static seal plate, the second seal plate, and the second static seal includes a plurality of second inner walls that define the second fluid passageway and a third fluid passageway between the louver plate and the second housing section.
10. The thermal management control valve of claim 9, wherein a rotational orientation of the first seal assembly about a center axis is different than a rotational orientation of the second seal assembly.
11. The thermal management control valve of claim 1, wherein the drive shaft includes a threaded portion.
12. The thermal management control valve of claim 11, wherein the threaded portion of the drive shaft is rotationally coupled to gear teeth arranged around a periphery of the louver plate.
13. The thermal management control valve of claim 1, further comprising a second louver plate assembly enclosed within the housing assembly, wherein the second louver plate assembly defines a plurality of third passageways between a second louver plate and the first housing section, and a plurality of fourth passageways between the second louver plate and the second housing section.
14. The thermal management control valve of claim 1, wherein selective rotation of the louver plate proportionally opens or closes the one or more of the plurality of first fluid passageways.
15. The thermal management control valve of claim 1, wherein each of the plurality of first fluid passageways and the second fluid passageway is in fluid communication with a fluid port arranged on the housing assembly.
16. A thermal management control valve comprising: a housing assembly including a first housing section and a second housing section, wherein the first housing section includes a plurality of first fluid ports and the second housing section includes a plurality of second fluid ports; a rotary actuator coupled to a drive shaft that extends into the housing assembly; and a louver plate assembly enclosed within the housing assembly and including: a louver plate including a louver cutout extending through the louver plate, wherein the louver plate is coupled to the drive shaft so that selective rotation of the drive shaft is configured to rotate the louver plate; a first seal assembly sealed between the louver plate and each of the plurality of first fluid ports; and a second seal assembly sealed between the louver plate and each of the plurality of second fluid ports, wherein selective rotation of the louver plate rotationally aligns the louver cutout with one or more of the plurality of first fluid ports and one or more of the plurality of second fluid ports to provide fluid communication between the one or more of the plurality of first fluid ports and the one or more of the plurality of second fluid ports.
17. The thermal management control valve of claim 16, wherein selective rotation of the louver plate proportionally opens or closes a flow area between the one or more of the plurality of first fluid ports and the one or more of the plurality of second fluid ports.
18. The thermal management control valve of claim 16, wherein a rotational orientation of the first seal assembly about a center axis is different than a rotational orientation of the second seal

assembly.

19. A thermal management control valve comprising: a housing assembly including a first housing section and a second housing section; a rotary actuator coupled to a drive shaft that extends into the housing assembly; a first louver plate assembly enclosed within the housing assembly and including a first louver plate arranged about a first central axis; and a second louver plate assembly enclosed within the housing assembly and including a second louver plate arranged about a second central axis that is spaced from the first central axis, wherein the first louver plate assembly and the second louver plate assembly both include a first seal plate assembly and a second seal plate assembly, wherein the first seal plate assembly of the first louver plate assembly is sealed between the first louver plate and the first housing section, and the second seal plate assembly of the first louver plate assembly is sealed between the first louver plate and the second housing section, wherein the first seal plate assembly of the second louver plate assembly is sealed between the second louver plate and the first housing section, and the second seal plate assembly of the second louver plate assembly is sealed between the second louver plate and the second housing section, and wherein selective rotation of the drive shaft is configured to rotate the first louver plate and the second louver plate in opposing directions.

20. The thermal management control valve of claim 19, wherein the first louver plate includes a first louver cutout and the second louver plate includes a second louver cutout, and wherein selective rotation of the first louver plate and the second louver plate rotationally aligns the first louver cutout with one or more of a plurality of first fluid passageways, and aligns the second louver cutout with one or more of a plurality of second fluid passageways.